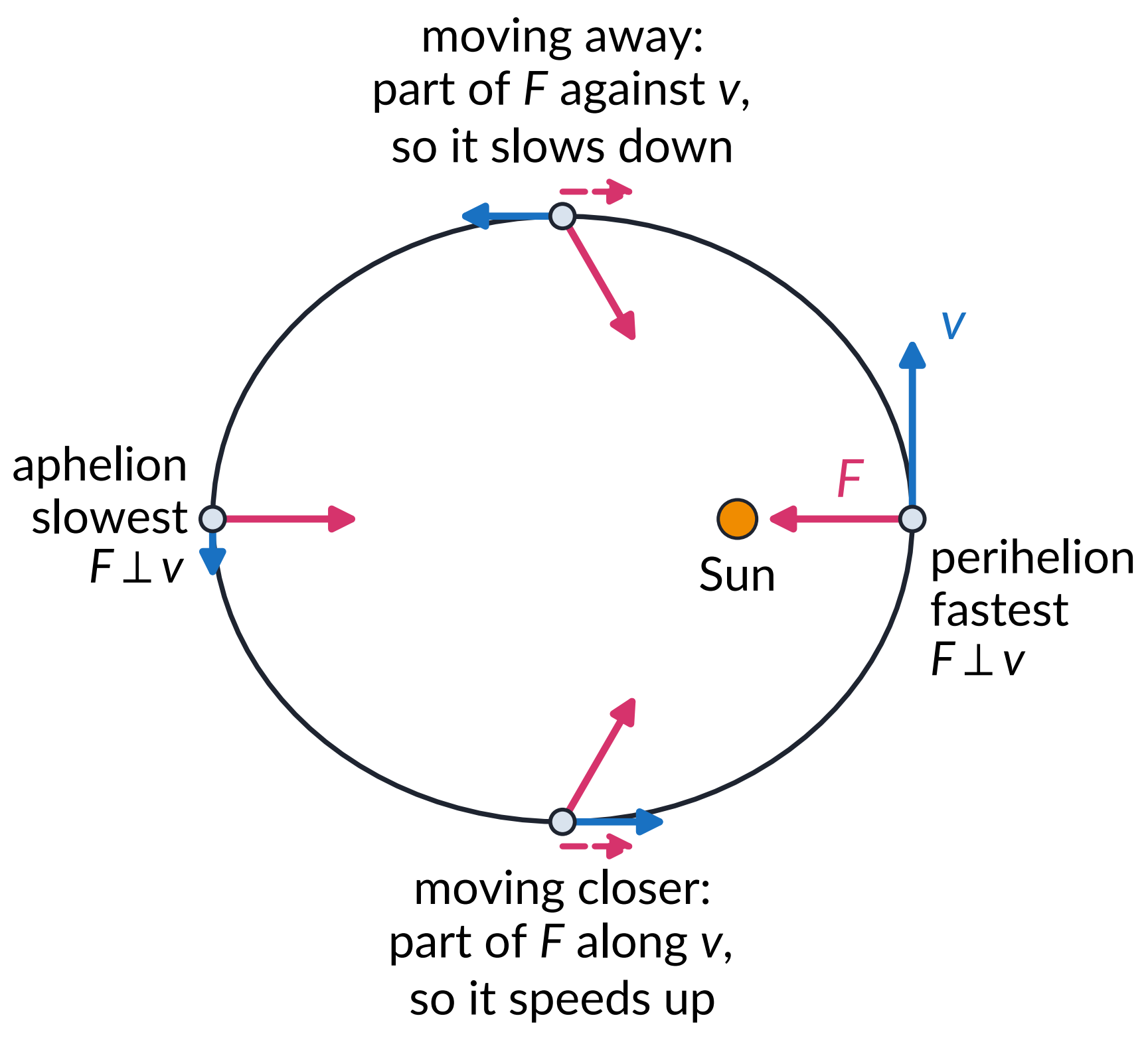
moving away:
part of F against v,
so it slows down
v
F
aphelion
slowest
F ⊥ v
Sun
perihelion
fastest
F ⊥ v
moving closer:
part of F along v,
so it speeds up